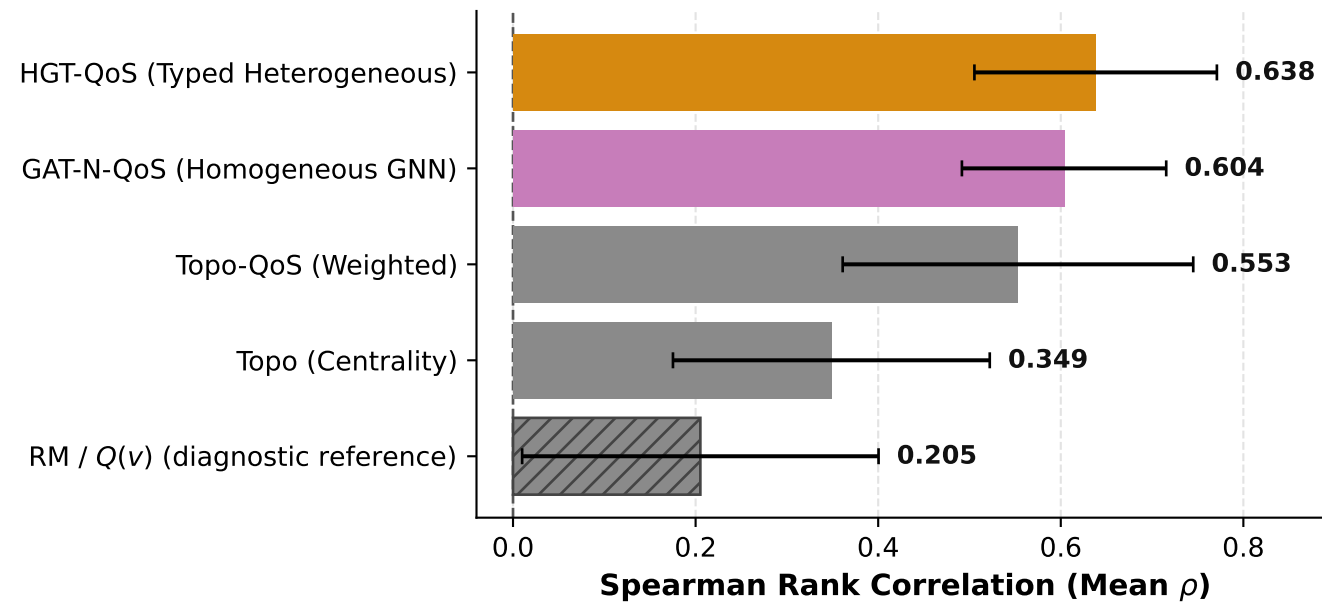
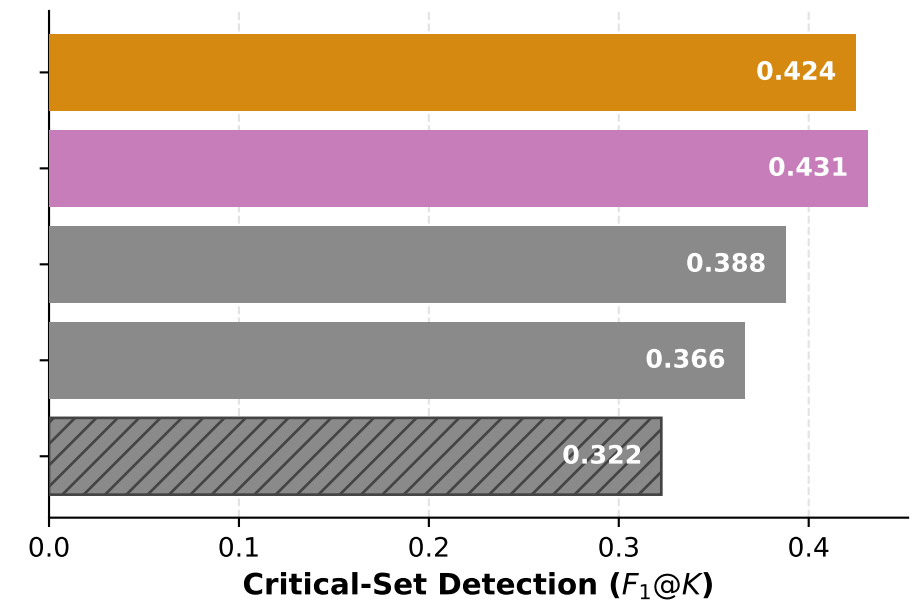


Software-as-a-Graph (SaG) Pre-Deployment Reliability Evaluation — application population

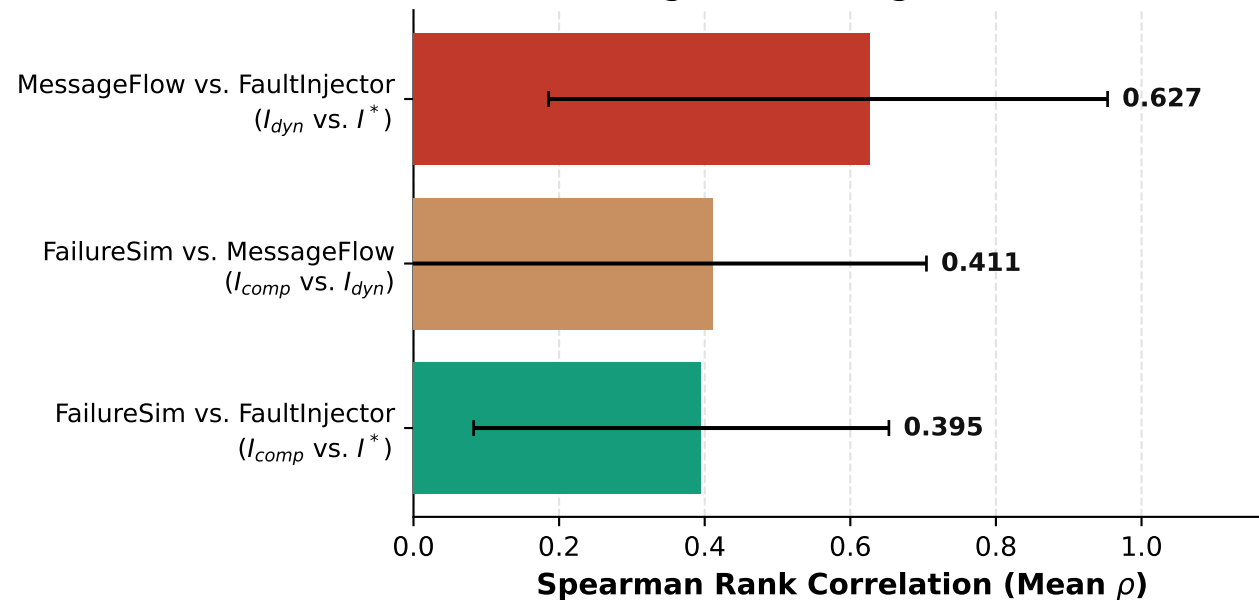
A. Out-of-Distribution Generalization (Leave-One-Scenario-Out ρ)



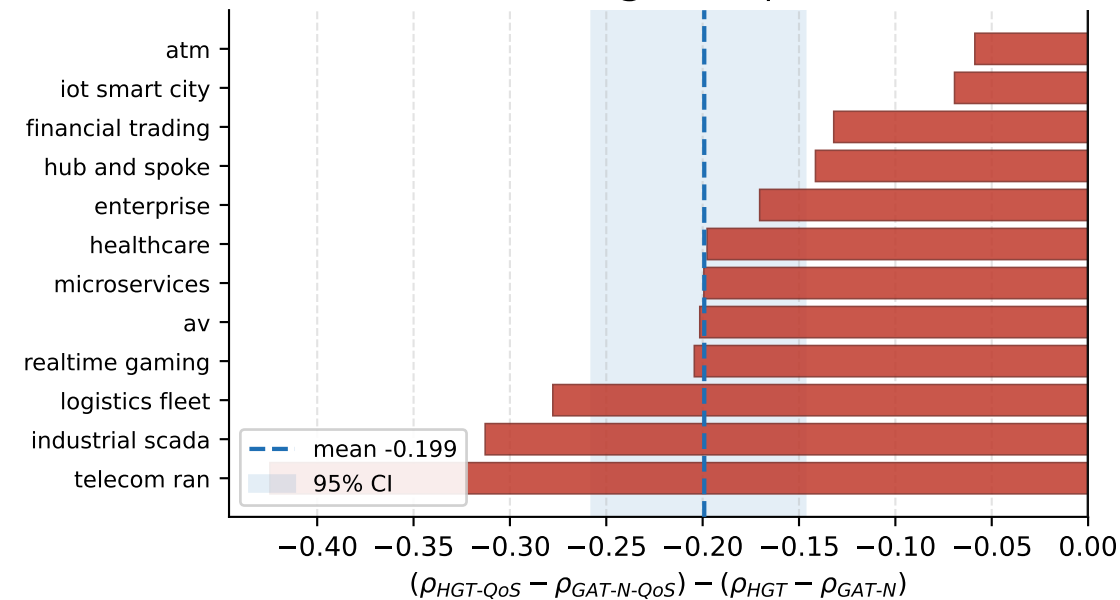
B. Top-K Critical Component Search ($F_1@K$, $K = \text{top } 20\%$)



C. Simulation Oracle Agreement (Rank Convergence & Range Across Scenarios)



D. Typing $\times$ QoS Interaction (12/12 folds negative, $p = 0.0005$)



Mean top-K Jaccard overlap: 26.1%–37.3% (chance = 11.1%)
Read against the labeler's own seed-to-seed floor: 37%–100%